DESIGN AND EVALUATION OF A STEAM COOLED FAST BREEDER REACTOR OF 1000 MW(e)*

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1. Introduction

At the end of 1964 the Karlsruhe group completed a relatively detailed design study of a 1000 MW(e) sodium cooled fast breeder reactor called Na 1.[1] This design was used as the basis of a subsequent system analysis[2] which showed the dynamic behaviour and the safety features as well as the effect on costs and safety of a number of design variations. Now a second design study (Na 2) is under way which will give the basic specifications for the prototype reactor of 300 MW(e).

In the same general pattern a design study of a 1000 MW(e) steam cooled fast breeder reactor (D 1) has now been completed. Some of the results of this design study will be presented in this paper. According to the principle mentioned above, we do not claim this design to be the optimum of all possible steam cooled fast reactors. We thought it to be more valuable to study first one reference design in more detail with the purpose of finding the physics and engineering problems typical of steam cooling. The more detailed optimization will then be made during the system analysis. This procedure has already shown a number of interesting results which allowed us to gain more insight into the key problems.

According to the German fast breeder programme, after completion of the second designs, Na 2 and D 2, the responsibility for the more detailed design and construction of the prototype plants for both coolants will be taken over by two large industrial groups. In this way it is hoped to keep the target dates.

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2. General Features of the H_2O-steam Cooled Fast Breeder Reactor

A number of studies have been published on steam cooled fast reactors.[3-6] Starting from these, Karlsruhe has hoped to achieve lower capital costs relative to sodium and, consequently, lower power costs. This seemed to be attainable by the use of approved components from modern conventional steam power plants, the application of the direct cycle proved in BWR's and the possible evolution from light water reactor technology. Besides, it seemed to be of general value to have an alternative solution in the case of unforeseen difficulties with sodium.

During the course of the work some new viewpoints came to light in respect of steam cooling.

It is felt that the steam cooled fast reactor is an attractive improvement of the light water reactor through certain features of the fast breeder, especially the cheaper fuel cycle; but it must be remembered that its breeding potential is lower than for sodium cooling. This will be shown in more detail in the following paragraphs.

Breeding Ratio

The breeding ratio of a light water steam cooled fast reactor, in the range of 150–180 atm pressure, will be lower than with the fast reactors with sodium cooling. This is for two reasons:

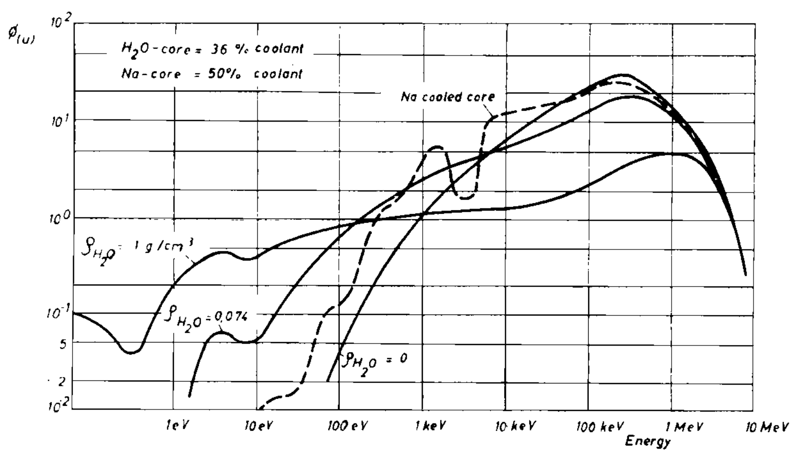

FIG. 1. Neutron flux spectrum.

(a) The energy spectrum is softer, as shown in Fig. 1. This results in somewhat smaller η-values and amplifies the effect of fission product poisoning.

(b) The temperatures of the structural material in general will be higher. This is especially true of the canning. Therefore, more structural material or a special high temperature material must be used to withstand the loads, e.g. Inconel 625. Also Inconel 625 is required because of corrosion. The content of 9% molybdenum and 4% niobium or tantalum in this material is unfavourable for breeding.

However, if one tries to compensate the influence of the softer energy spectrum by making the coolant fraction in the core as small as possible in respect of the design and tolerance problems, e.g. 32% on the average, the total breeding ratio of a 1000 MW(e) steam cooled fast reactor of medium burnup is still approx. 0·1 lower than in a sodium cooled reactor of the same burnup and with a coolant fraction of 50%. For sodium cooling, breeding can even be raised by the use of carbide fuel. In that case the difference amounts to about 0·25. It should be borne in mind that the use of other steam pressures or heavy water will change these figures.

Rod Power and Rating

In the sodium cooled reactor the rod power is limited by the melting point and the thermal conductivity of the fuel material. For steam cooling, the canning temperature and the heat transfer to the coolant are also limiting values. This results in a larger influence of the hot channel factors for steam cooling. The deviation of a temperature difference from the design value is proportional to this difference and to the hot channel factor. The larger temperature difference between can and coolant results in a higher canning temperature at the hot spot. In addition, two hot channel factors are higher than for sodium:

(a) the relative deviations in coolant channel dimensions are larger because of the smaller coolant fraction;
(b) the heat transfer number depends more strongly on the coolant flow and, therefore, on the channel diameter.

The maximum can temperature is defined by mechanical considerations. Here, the fresh condition is important where the canning is still under external pressure. Detailed studies were made of creep buckling under these conditions and it was found to be the dominant effect. For Inconel 625 and a typical structural material content in the core, a maximum temperature of approximately 760°C on the inside of the can and at the hot spot seems to be a limiting value. For this a remarkable internal inert gas pressure had to be assumed.

Figure 2 shows the life of a cylindrical tube related to temperature for different external pressures. Note the strong dependence on temperature. The results of a number of thermodynamical calculations* for the hot spot temperature of 760°C on the canning inside are shown in Table 1. The coolant pressure = 170 atm, the coolant inlet temperature = 365°C, the coolant outlet temperature = 540°C, the pressure drop = 6·6 atm and, therefore, the plant efficiency

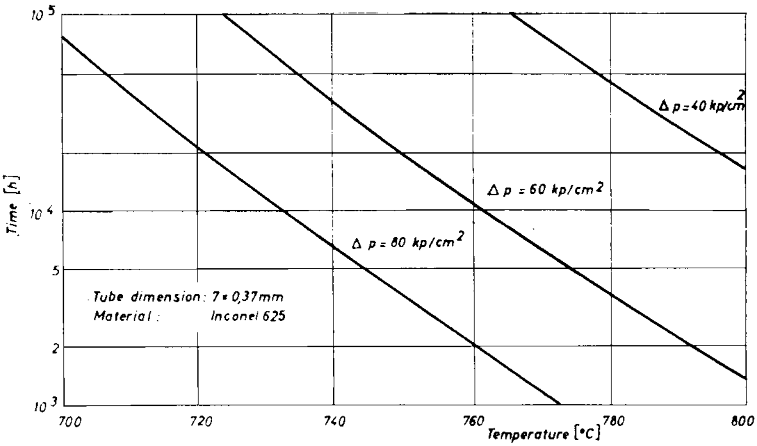

FIG. 2. Time to collapse related to temperature for cylindrical tubes and different external overpressure.

TABLE 1. SEVERAL THERMODYNAMICAL RESULTS

	Core 1	Core 2	Core 3	Core 4	Core 5
Rod diameter (mm)	6·4	7	8	7	7
Core height (cm)	132	144	167	75	75
Core diameter (cm)	260	260	260	320	247
Power (MW(e))	1000	1000	1000	1000	600
Maximum rod power† (W/cm)	403	443	497	483	483
Average rating (MW/kg Pu)	0·78	0·73	0·63	0·77	0·74
Pressure drop (atm)	6·6	6·6	6·7	6·0	6·0
Coolant fraction α (%)	32	32	32	22	22

† Hot channel.

is also constant. Power shape factors of $\psi_{rad} = 0{\cdot}86$ and $\psi_{ax} = 0{\cdot}74$ were used. Cores 1 to 3 show the maximum rod power for different rod diameters and the attainable rating for 1000 MW(e), a coolant fraction of 32%, and a core diameter of 260 cm.

The rod power increases linearly with the diameter because the heat flux

* The relations for heat transfer and pressure drop are based on measurements of R. Moeller, Institute for Reactor Components, Nuclear Research Centre. Karlsruhe.

stays nearly constant. By contrast, the rating decreases because a larger critical mass is required with a larger core volume. These rod powers and ratings shown are relatively low compared to a sodium cooled reactor. However, these figures could be raised to a certain extent, e.g. by the choice of a smaller value for the outlet temperature; but this would decrease efficiency. The system analysis has to show if the power generation costs can be lowered by going in this direction.

Reactivity Coefficient of Coolant Density

Figure 3 shows the dependence of k_{eff} on the coolant density for different burnup and a typical cylindrical 1000 MW(e) core. Owing to the effect of

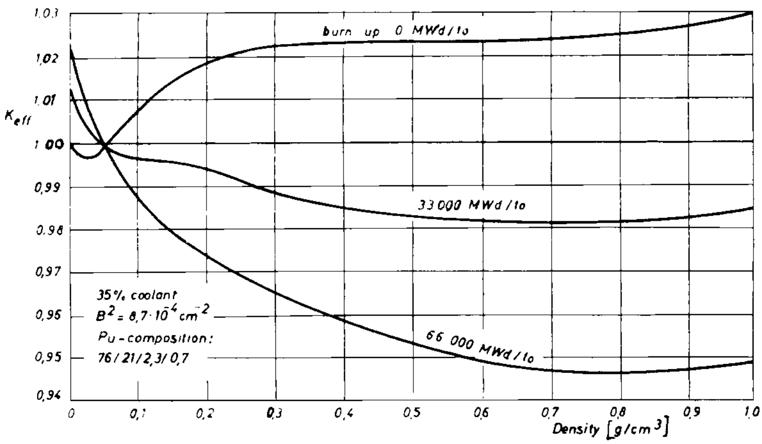

FIG. 3. k_{eff} related to coolant density for different burnup.

fission products the power coefficient of coolant density becomes more and more positive. This is of small importance for reactivity disturbances because of the very large negative Doppler coefficient of H_2O-steam cooled reactors (see also Table 3).

Nevertheless, the power coefficient stays relatively constant over a longer period of operation since practically only the shim absorbers will be replaced by fission products.

The total loss of coolant will result in a relatively high positive reactivity. Under the assumption that even for a large leakage the total loss of coolant will need some seconds, the produced reactivity ramp will amount to less than $10 per second. Therefore, the total energy production during an excursion of this type will not be excessive.

Economy

The lower breeding ratio of the light water steam cooled reactor gives a larger doubling time. Therefore, the sodium system leads to a better strategy in the optimum use of fuel reserves. Table 2 shows a first comparative view on the power generation costs. These figures should be interpreted as a tendency which must be proved by later detailed evaluations.

TABLE 2. COMPARISON OF POWER COSTS
(mills/kWh)

	LWR	D 1	Na 1
Fuel cycle	1·9	1·2	0·875
Capital costs	2·56	2·90	3·13
Operation costs	0·24	0·24	0·24
Total	4·7	4·34	4·25

The costs for the light water reactor (LWR) are based on evaluations of German industry. Compared with these figures, the steam cooled D 1 has higher capital costs owing to some additional circuit components and to the higher coolant pressure. On the other hand, the turbine will be less expensive. By contrast, the fuel cycle costs are definitely lower for the fast reactor. Their savings still allow for more attractive total power generation costs. Even a certain increase in fuel cycle costs would be tolerable so that, for example, the burnup could be reduced from 100,000 MWd/t to some 55,000 MWd/t. This would improve the breeding ratio and the coolant density coefficient and simplify the design of the fuel elements to a certain extent.

It is a fact, however, that the attractive fuel costs of the sodium cooled breeder could not be reached with a steam cooled system.

As shown in Table 1, the rating increases with a smaller rod diameter. Some improvements of the fuel cycle costs for the steam cooled reactor seem possible in this way if the difficulties combined with small rod diameters and a close lattice could be solved, and the fabrication costs of the fuel elements would not increase too much.

However, all considerations for cores 1 to 3 are strongly influenced by the core diameter and the coolant fraction in-core. If we could neglect this limitation, a more favourable rating would also be possible with a smaller core height, as shown in core 4. In this example also the pressure drop is some 10% lower and, correspondingly, the efficiency is improved. Unfortunately, the core diameter of core 4 is very large. Almost no radial blanket should be used in order to avoid immense trouble with the pressure vessel.

Therefore, the power output must be decreased in this case to perhaps 600 MW(e), as shown by core 5.

The coolant fraction of cores 4 and 5 is extremely small and leads to tolerance problems. An improvement would be possible perhaps by using

TABLE 3. MAIN PARAMETERS OF THE REFERENCE DESIGN D 1

Total power	2517 MW(th)	
Net power output	1000 MW(e)	
Overall efficiency	39·7%	
Coolant	H_2O steam	
Coolant pressure	170 kg/cm^2	
Inlet temperature	365°C	
Outlet temperature	540°C	
Core height H	151 cm	
Core diameter D	263 cm	
H/D-ratio	0·575	
Number of core zones	2	
Fuel rod outside diameter	7 mm	
Pitch in hexagonal lattice	8·15 mm	
Fuel fraction in core	46·5%	
Coolant fraction in core	31·6%	
Structural material fraction in core	21·9%	
Blanket thickness	38 cm	
Max. rod power with hot channel allowance	415 W/cm	
Max. canning temperature with hot spot allowance	747°C	
Mean power density	292 kW/litre	
Plutonium composition	74·0% Pu 239 22·7% Pu 240 2·3% Pu 241 1·0% Pu 242	
Burnup max. (axial average)	55,000	/100,000 MWd/t U+Pu
Mean power/max. power = $\psi = \psi_{rad} \cdot \psi_{ax}$	0·63	/0·62
Mean specific power	0·73	/0·71 MW/kg Pu 239+241
Fissile loading (Pu 239+241)	3274	/3372 kg Pu
Enrichment:		
inner zone	10·1	/10·6 $^{a}/_{o}$
outer zone	12·4	/13·0 $^{a}/_{o}$
Internal breeding ratio	0·95	/0·91
Total breeding ratio	1·15	/1·11
Doppler coefficient (900°K)	$-1{\cdot}75 \times 10^{-5}$	$/-1{\cdot}6 \times 10^{-5}\,°C^{-1}$
Steam density coefficient	$-0{\cdot}27 \times 10^{-3}$	$/-0{\cdot}38 \times 10^{-3}(kg/m^3)^{-1}$
Reactivity worth due total loss of H_2O	$3{\cdot}6 \times 10^{-2}$	$/4{\cdot}9 \times 10^{-2}$

turbulence promoters on the cladding surface, because in general this requires a larger channel diameter. However, this would again increase the core volume.

The detailed optimization between breeding, rating, reactor costs, and efficiency will be made during the system analysis. Preliminary results show, however, that light water steam cooling becomes somewhat more favourable for smaller power stations. This tendency seems to be true for the fuel costs as well as for the capital costs compared with sodium cooled systems. At present we cannot say at what power level a break-even point for the power costs could be expected. Also in capital costs another break-even point can be expected since decrease of specific capital costs is more pronounced for sodium than for steam because of the low pressure. Anyway, this information could only be drawn from optimized detailed designs for both coolants based on the same safety requirements and on the same technological development standard. Moreover, it would require perhaps the construction and operation of both systems under similar conditions.

However, steam cooling may still be attractive and find a good market. Hence, both systems will not exclude each other but are perhaps supplementary in important features and are complementary solutions.

3. Reference Design D 1—Basic Decisions

With respect to the foregoing and other considerations the following parameters were fixed.

Fuel and Fuel Element

A UO_2/PuO_2 mixture is used as the fuel; an external pin diameter of 7 mm was selected as a compromise between high rating and good mechanical stability. The cladding material is Inconel 625 because of its good high temperature strength and corrosion behaviour. Spiral fins as integral parts of the cans are used as spacers. To prevent creep deformation, the rods are filled with noble gas at a pressure of 120 atm. Fission gases are stored in a plenum, since venting to the coolant must be excluded for steam cooled reactors, especially with a direct cycle.

Referring to the foregoing considerations, mainly the creep deformation problems, the maximum rod power was set at approximately 415 W/cm. With this figure a rating of approximately 0·73 MW/kg fissionable material will be reached.

Shape of Reactor Core

For the core a cylindrical shape with two enrichment zones of equal volume was chosen. With respect to the pressure vessel size the outer diameter was set at some 260 cm. This led to a height to diameter ratio of nearly 0·6.

Coolant and Coolant Pressure

Light water steam is used as the coolant. Heavy water or a mixture of heavy and light water was excluded because we feel that the better breeding ratio does not compensate for the greater difficulties during refuelling and for the increased tritium problems. An average coolant pressure of 170 atm and a suitable steam outlet temperature of 540°C was chosen. This gives a relatively good heat transfer, good overall efficiency and allows the use of a modern conventional steam turbine. Besides, an extrapolation to higher and lower pressures during the following system analysis seems possible.

Direction of Flow in the Core

The cooling steam flows in an upward direction through the radial blanket and inner thermal shield and then downwards through the core. In this way, any residual moisture will be evaporated before reaching the core, and a special holddown device for the fuel subassembly is not required. Also, for partial core melting the compactness of the core is reduced.

Control Rods

The control rods are on subassembly positions. This allowed for a smaller number of rods and uniform subassembly cross-sections for the whole core. The control rod drives are mounted on top of the vessel head and are connected to a grid plate above the core to prevent a control rod blow-out in case of a nozzle break. Therefore, the driving shafts do not disturb the lower outlet plenum for the hot steam and the drives do not occupy the space below the pressure vessel. This allows a simple and novel arrangement of the reactor internals and the coolant pipe connexions.

Refuelling System

Refuelling will be done under water. This allows for quick operation, gives high performance, and is one of the important advantages of the light water reactors which should be retained. For this purpose the design will make provisions for easy flooding of the reactor. This flooding possibility is also an advantage with respect to decay heat removal after some accidents.

Coolant System and Power Plant

The reactor coolant system uses the Loeffler cycle with spray type steam generators. The plant will operate in direct cycle. This allows for lower capital costs. With respect to component size and safety considerations, six independent Loeffler circuits were used. For decay heat removal and some other duties two subsidiary coolant circuits with surface type steam generators are provided. To reduce capital costs still more and to reach maximum access to the plant during operation and after an accident only such components will be installed in the containment vessel as are necessary to fulfil the safety requirements.

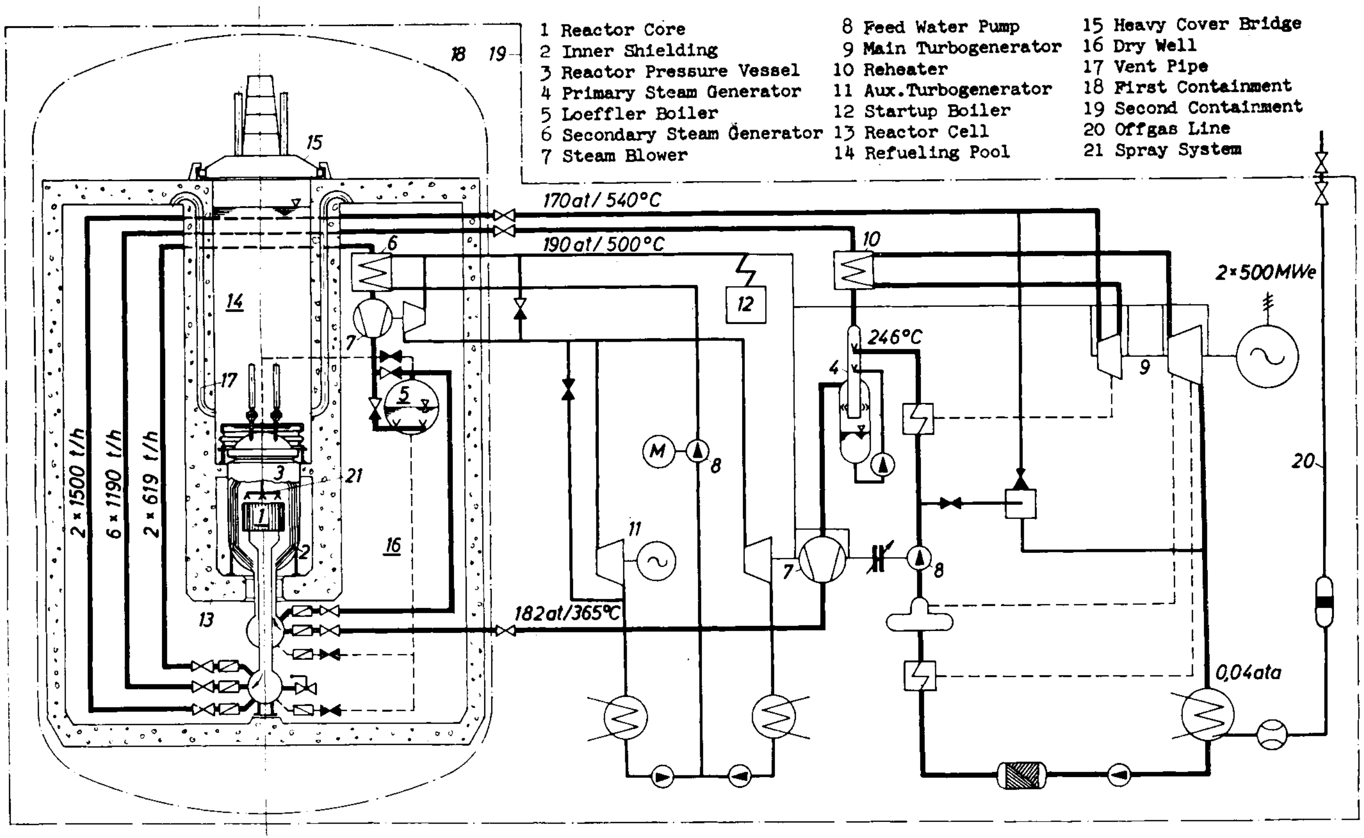

FIG. 4. Simplified flow diagram and containment system.

Containment

The containment system was designed in agreement with the following design criteria. All vessels, containers, or rooms which could contain fission products or plutonium aerosols above atmospheric pressures must be enclosed by a second gastight envelope and protected by an inner blast shield against any damage. To lower the maximum containment pressure a pressure suppression system has been used.

4. Reference Design D 1—Outline Description

A simplified flow diagram and the containment system is shown in Fig. 4. Six main Loeffler circuits equipped with spray type steam generators and two smaller subsidiary cooling loops with surface type steam generators are connected to the reactor. Two relatively small standard Loeffler boilers are also interconnected to these loops. In addition two steam lines lead from the reactor to the two main turbogenerators feeding in the working steam at 170 atm pressure and 540°C. After expansion in the HP-cylinders of the turbines, this steam is reheated in surface type reheaters, which are part of the main Loeffler circuits. With this system 1000 MW(e) power is produced from 2517 MW heat generated in the reactor The overall efficiency amounts to 39·7%.

During reactor operation the two subsidiary loops are in continuous operation. The non-radioactive secondary steam generated in these systems is used to feed the driving turbines of the steam blowers and other auxiliary equipment of the plant. Moreover, it is used as a non-radioactive steam barrier in the labyrinth type seals of the machines run on radioactive steam.

The two Loeffler boilers generate only a small quantity of steam during normal operation. Their main purpose is to act as buffer and storage in case of power transients, equipment failures, or arising leakages. Besides this, the volume of water in the two Loeffler boilers, which are located at a higher level than the reactor, is used to flood the reactor for the refuelling procedure. After shutdown the reactor and a first steam cooling period this process of flooding has to be carried out at a relatively high pressure because of the decay heat generation. The flooded reactor then is cooled by the use of a pressurized water system which is not shown on Fig. 4. Simultaneously, the pressure and temperature are reduced to the conditions of the refuelling process in conformity with the temperature changes admissible for the reactor components. At the same time it is possible to connect a primary water purification and degasification system. The whole depressurization and flooding process is shown graphically in Fig. 5. Draining the flooded reactor follows roughly the same sequence but in opposite directions. The rapid increase in reactivity (see Fig. 3) during this process requires some safety precautions.

The layout of the plant components and of the protective containment walls was dictated by safety considerations. The core is enclosed in a heavy steel pressure vessel; the neutron and γ-shield in the vessel acts as a first blast shield. The pressure vessel is located in a reinforced concrete reactor cell designed to withstand the pressure arising in the maximum credible accident. The walls of this cell also act as a biological shield. The upper part of this cell is filled with water thus constituting a refuelling pool. Final protection in this sense is provided by a heavy movable cover bridge.

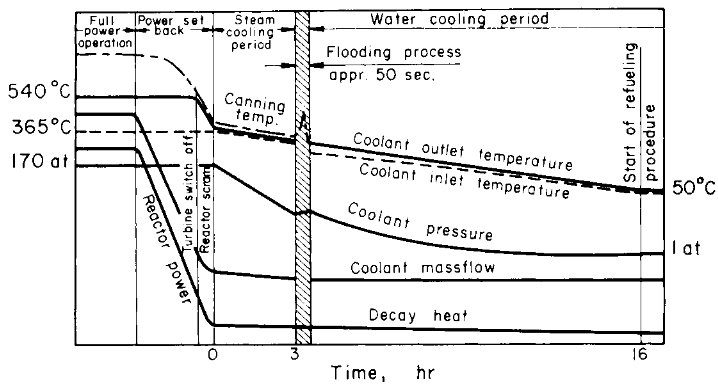

FIG. 5. Sequence during the depressurization and flooding process.

All coolant pipes with their valves, the pressurized water cooling system, and the purification circuits as well as the equipment of the two subsidiary coolant loops including the two Loeffler boilers are located below and around the reactor cell in a dry well formed by concrete walls. Any increase in pressure in this volume, which may be caused by a reactor accident or any leakage, is dissipated through ventpipes into the refuelling pool and ultimately to the pressure resistant first containment. This first container, the six main Loeffler circuits, and the two main turbogenerators in turn are enclosed in a second gastight envelope. Here, a small negative pressure is maintained. It can be expected that this system will control all credible accident conditions as well as handle and discharge in a controlled manner any potential release of radioactivity.

Under shutdown conditions all penetrations between the first and the second containment, which may contain radioactivity, are closed. For this purpose one quick acting check or overload stopvalve, respectively (adjacent to the reactor pressure vessel) and two independent normal stopvalves are installed in each pipe line. In this case the decay heat is removed by the two subsidiary coolant loops or, under flooded conditions, by the water cooling system.

Also during the whole starting procedure the penetrations remain closed until the reactor power exceeds about 10%. The plant operates during that period as a dual cycle system with the two auxiliary turbines and their condensers.

Full power production occurs in direct cycle similar to the well known BWR's. Nearly 10,000 t/h saturated steam are generated at maximum load in the six spray type steam generators and pumped to the reactor. Some additional 1200 t/h coolant steam is delivered from the two subsidiary

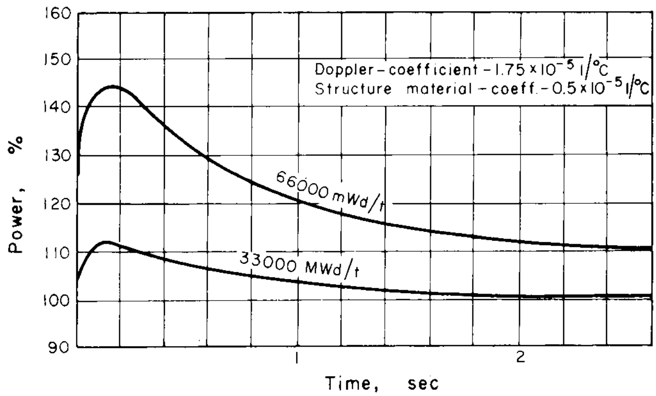

FIG. 6. Power related to time for different burnup rates after a sudden coolant density decrease of 0·006 g/cm³.

systems. In the reactor this steam is superheated to 540°C. The steam flow coming from the reactor is then divided into three paths. Approximately 3000 t/h go to the two main turbines, approximately 7000 t/h to the six Loeffler circuits and the rest to the two subsidiary systems.

Any sudden positive power step by the two main turbogenerators of an order permissible in modern conventional power stations will not result in an unacceptable decrease in steam pressure or in an undue increase in power, respectively. The water reservoir in the bottom sections of the main steam generators, where the water separated out of the saturated steam is collected, has a stabilizing effect in this case. It keeps the steam pressure nearly constant at this point of the circuit so that any pressure change at the reactor outlet will strongly influence the relatively small pressure difference to this point. A decrease in this pressure difference, however, also decreases the amount of steam supplied to the steam generators. Consequently, more steam is available for the main turbogenerators. Additional steam required during this transient is generated by the water reservoir of the steam generators. A sudden decrease in power could easily be controlled by a bypass system to the main turbines.

A first dynamic analysis has been made for several disturbances in the coolant circuit. In general the Doppler effect will stabilize the conditions after some overswing. As example, Fig. 6 shows the reactor power as function of time for different burnups after a relatively large sudden coolant density decrease of 0·006 g/cm^3. This value corresponds nearly to a pressure decrease of 10 atm or a temperature increase of 25°C, respectively. We feel now that the steam cooled fast breeder reactor will not give extremely difficult control problems.

However, it is possible that a more detailed analysis of the dynamic behaviour of the plant shows that the water volume in the main steam generators is no necessary prerequisite. In that case it would be possible also to use cheaper types of steam generators, perhaps those in which the steam produced is still slightly superheated. Moreover, operation with supercritical steam pressure would be possible.

In a similar way the two Loeffler boilers stabilize the pressure on the reactor inlet side. This is important if a leak develops at this section of the cooling system. The Loeffler boilers then prevent unduly rapid decrease of the steam flow passing the core and of the pressure in the cooling system, so that special safety measures can be taken. Leaks located in the cooling systems may be isolated from the rest of the reactor system by closing of the proper valves. Leaks of the reactor vessel, however, could not be isolated; they will ultimately lead to a depressurized reactor. In this case a water spray system will go into operation, by which the coolant steam flowing into the core would be mixed with water spray, thereby remarkably increasing its cooling efficiency so as to prevent meltdown of the core. In addition emergency flooding of the reactor will be carried out as early as possible. If the leak is below core level additional water is fed into the cooling system and the whole dry well is flooded with cooling water from the main condensers until the core will stay below water level. This latter method of flooding is also carried out after a maximum credible accident. It will also allow the decay heat to dissipate for unlimited periods of time also out of a completely destroyed reactor.

Another abnormal operating condition or accident, respectively, which has to be regarded is an increase in radioactivity in the turbine steam, e.g. as a result of a fuel element leakage, cladding failure, or fuel damage. If the increase is relatively slow the defective subassembly can be identified by a fission product detection system which is in continuous operation. When a certain activity level has been reached the operation and power level of the plant is then dictated by the dose rate measured at the turbines and the offgas lines until the defective subassembly is replaced.

Any heavy and sudden increase in radioactivity, however, will require immediate safety measures. In that case the reactor is scrammed and all penetrations in the first containment which carry radioactive steam are closed. The checkvalves and the overload stopvalves are, therefore, designed for a short closing time. However, operation of these valves alone is not sufficient to guarantee complete leak tightness and reliability. Therefore, in that case the two independent normal stopvalves in each duct are also closed. The system analysis may show perhaps that a controlled fast power setback is also adequate in this case. This would avoid excessive thermal stresses in the cooling circuit.

During and following this transition from *open cycle power production* to *closed cycle shutdown condition* the two subsidiary cooling circuits are not

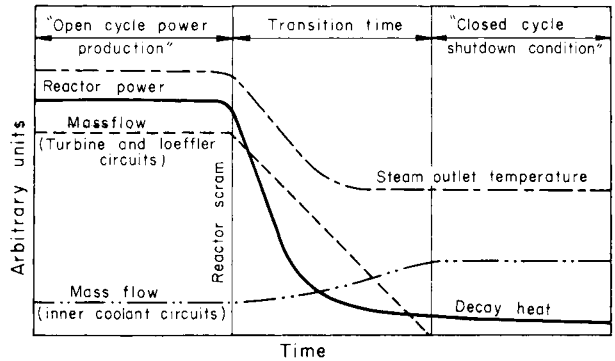

FIG. 7. Transition from *open cycle power production* to *closed cycle shutdown condition.*

affected. Hence, no additional steps are necessary to safeguard further removal of the decay heat.

The behaviour of the cooling system during this transition is shown qualitatively on Fig. 7. Interesting is the remarkable mass flow increase in the subsidiary coolant loops as a result of the decrease in reactor pressure drop. Therefore, only one subsidiary loop with one steam blower operating is adequate for the decay heat removal as long as the steam pressure remains close to the operating level.

In this connexion it must be mentioned that the four steam circulators within the first containment (each loop has two blowers) on which the decay heat removal essentially depends are of a particularly simple design, so that a high degree of reliability may be expected. All other important equipment for the decay heat removal system, e.g. feed water pumps, condensers, etc., is located outside the first containment and is not radioactive, so that it remains accessible in any case. Leakages from the radioactive reactor cooling circuit to the non-radioactive secondary system are prevented because the secondary pressure is greater than the reactor coolant pressure by some 20 atm. A conventionally fired startup boiler is also available as a standby energy source to drive the steam blowers.

Figure 8 shows a longitudinal section through the reactor. The cylindrical

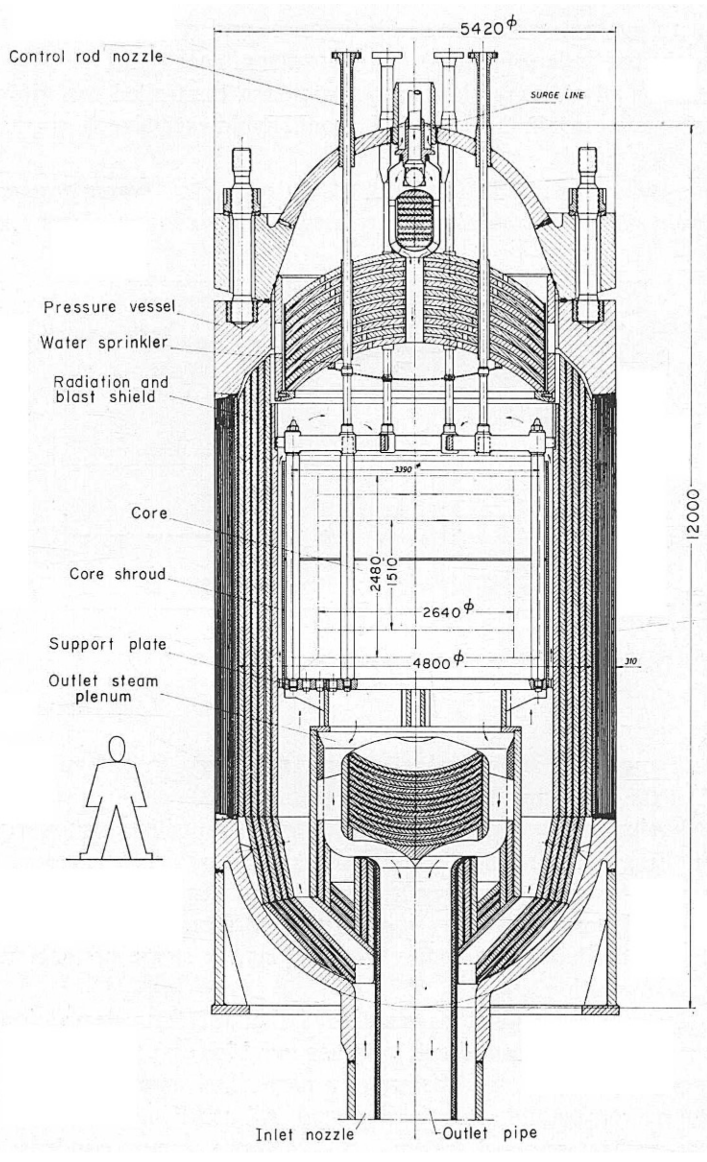

FIG. 8. A longitudinal section through the reactor.

core and blanket arrangement consists of 163 fuel subassemblies, 18 control subassemblies, and 120 blanket subassemblies of an identical hexagonal cross-section. The central core region contains the fuel in two enrichment zones, the outer region and the fertile material. In each fuel subassembly 469 fuel elements are installed.

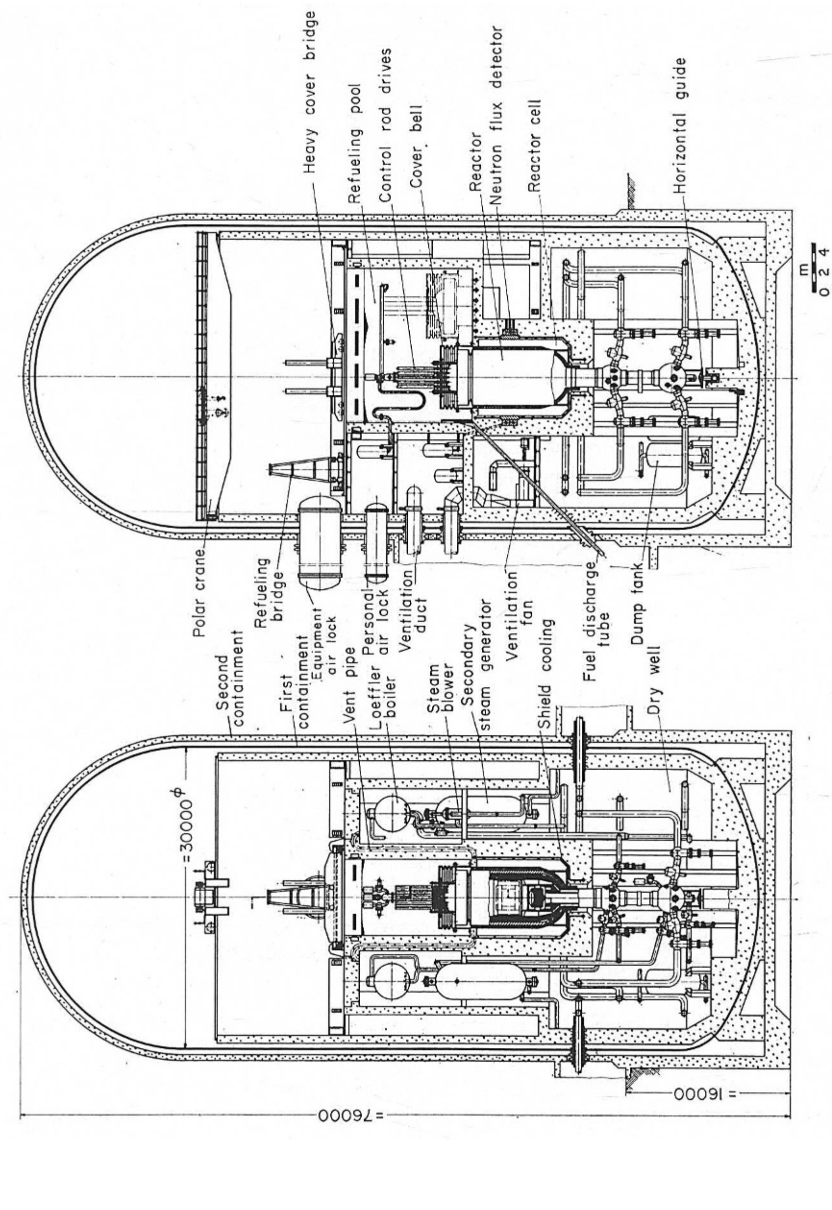

FIG. 9. Reactor building.

The whole core structure is supported by a bottom grid plate and held together by an external shroud. The cooling steam enters the reactor vessel from below through the annular channel of the big concentric stud, flowing first from the bottom to the top through the radial blanket and through the inner radiation shielding established between reactor core and pressure vessel wall. In this way it reaches the upper plenum chamber and flows down again into the core of the reactor. When it has been heated it leaves the reactor pressure vessel through the inner channel of the concentric stud at a temperature of 540°C. A surge pipe is attached on top of the pressure vessel which leads to the steam section of the Loeffler boilers. Normally, the inner opening is closed by a special device and only opened during the flooding process. Through a central hole in this device the water is delivered to the water sprinkler.

A remarkable feature shown in this drawing is the high expenditure for inner radiation shields necessary for a steam cooled fast reactor in order to keep exposure of the walls of the pressure vessel low. In this design the only shielding material used was steel. The thickness has been selected to make the exposure of the wall of the vessel over 30 years not exceed approximately 2×10^{19} neutrons/cm^2, which is a permissible value for ferritic material. Moreover, this shielding keeps the activation of the top flange low so that work can still be performed on it with the proper safety precautions.

The pressure vessel has relatively large dimensions. The inner diameter is 4·8 m, the height about 13 m, and the wall thickness of the cylindrical portion built in multi-layer construction is about 300 mm. The bulk weight of the vessel without installations will be some 540 tonnes. Owing to these dimensions it is necessary to weld the vessel on the site out of several prefabricated sections. But there are no fundamental obstacles in the way of this type of construction.

A longitudinal section through the reactor building is shown in Fig. 9. The reactor pressure vessel extends down right into the dry well forming a spherical header in this part. In this arrangement the stud connexions of the pipes and the penetration of the hot-steam line through the pressure vessel wall which is maintained at lower temperatures remain accessible. Moreover, it is possible to arrange the fast acting valves near the pressure vessel and install the pipes within the reactor building in a sufficiently flexible way. This way of pipe layout specially safeguards that, in an excursion of the reactor, the penetrations through the wall of the containment are not impaired.

For the refuelling process the pressure vessel head is lifted off under water and laterally disposed in the refuelling pool. Transport is carried out by means of a special movable heavy lifting bridge.

Refuelling is carried out in direct view by a second lighter movable bridge carrying all the necessary equipment. The spent fuel elements are discharged into a fuel element storage bay outside the containment through a shielded inclined downward transfer tube. The first containment is surrounded on the outside by an additional gastight concrete shield; this is the second containment.

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